Venkata Lakshmi Parimala Pasupuleti

ML Code Reviewer · Python · PyTorch · TensorFlow · scikit-learn · Model Evaluation · LLM-as-Judge · AI Training Data · MLOps · HuggingFace

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PROFESSIONAL SUMMARY

ML practitioner and AI systems engineer with hands-on experience in model training, evaluation, and production deployment across Python, PyTorch, TensorFlow, and scikit-learn. Fine-tuned a custom classification model using HuggingFace Transformers + PyTorch at GMU; built ML pipelines (Random Forest, Logistic Regression, Isolation Forest) on 100K+ record datasets; built AgentProbe, an open-source AI evaluation toolkit with LLM-as-judge scoring - directly adjacent to RLHF/AI training workflows. 8+ years Python production engineering. Strong written communication: delivered technical workshops to 600+ students, explaining ML concepts and code logic for varying skill levels. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

ML EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate · College of Engineering and Computing

- **Fine-tuned a custom classification model** using HuggingFace Transformers and PyTorch — designed the architecture, ran training, evaluated results, and deployed an inference API on Quart (async Python framework); presented model design, training methodology, and evaluation results to faculty.
- Built multi-source prediction models (Python, R) on WHO + UK Road Safety datasets across 190+ countries — R^2=0.977; full pipeline from data collection, feature engineering, and preprocessing through hypothesis testing, cross-validation, and model selection.
- Applied Isolation Forest anomaly detection across 3 research pipelines — surfaced data quality issues that invalidated assumptions before they propagated to model outputs; improved downstream model reliability.
- Designed Tableau, ggplot2, and Quarto dashboards; explained model findings and methodology to non-technical faculty audiences.

Independent Projects
2025 – 2026

ML Engineering & AI Evaluation

- **AgentProbe** (open-source): Built a 5-metric AI evaluation engine — correctness, groundedness, latency, token efficiency, hallucination rate — with LLM-as-judge scoring, golden datasets, CI-integrated regression harness, and model routing across OpenAI/Anthropic/Groq/Ollama. Directly adjacent to RLHF and AI training data workflows. github.com/venkatalakshmiparimala/AgentProbe
- **CrashClarity**: End-to-end ML pipeline on 104K+ accident records — Random Forest + Logistic Regression with feature engineering across weather, road, and temporal variables; 85%+ accuracy; deployed interactive dashboard on GitHub Pages.
- **Leasel**: NLP pipeline over 1,240 lease PDFs — TF-IDF + semantic retrieval, LLM reasoning, LLM-as-judge evaluation harness achieving 93.6% precision and 87.9% F1 across 10 compliance classification rules. github.com/venkatalakshmiparimala/Lease_complaince_rag

George Mason University
Aug 2025 – May 2026

Graduate Teaching Assistant

- Delivered technical workshops to 600+ students on Python programming, data structures, and database design — explaining code logic and problem-solving strategies clearly for varying skill levels; graded assignments with rubric-aligned feedback specifying exactly why an answer was correct or incorrect.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- 8+ years Python production engineering — data pipelines processing 1.6M+ records, optimized 200+ SQL queries (30% gain), microservices redesign (50% scalability improvement). Strong foundation in writing efficient, scalable, production-grade Python code and reviewing it for correctness and edge cases.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87 — coursework: machine learning, statistical modeling, data engineering, NLP

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

ML Frameworks: PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, XGBoost, Isolation Forest, Random Forest, Logistic Regression

ML Practice: Model Training, Fine-Tuning, Model Evaluation, Feature Engineering, Data Preprocessing, Anomaly Detection, Classification, Regression, Inference API Deployment, MLOps, LLMOps

AI Evaluation: LLM-as-Judge, Golden Datasets, Correctness Criteria, Hallucination Detection, Groundedness Verification, Regression Harnesses, CI-Integrated Quality Gates, RLHF-adjacent evaluation workflows

Languages & Tools: Python (8+ yrs), R, SQL, NumPy, Pandas, SciPy, NLTK, spaCy, BERT, Jupyter, Matplotlib, Seaborn, AWS (S3, EC2, SageMaker), Docker, Git